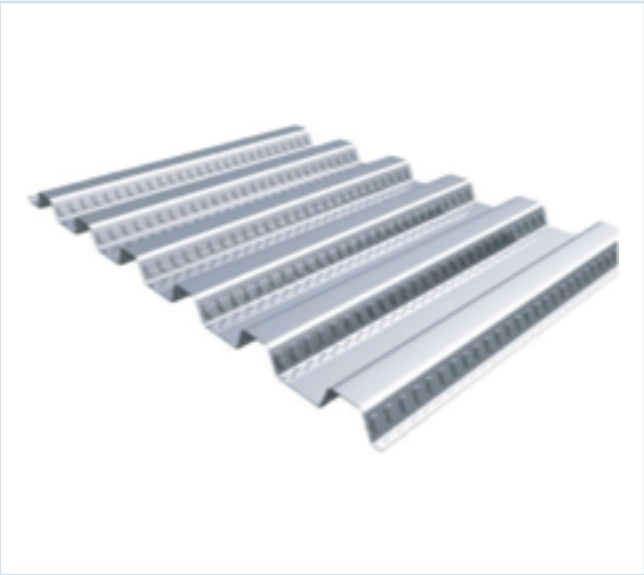


DECKING PRODUCT SUBMITTAL

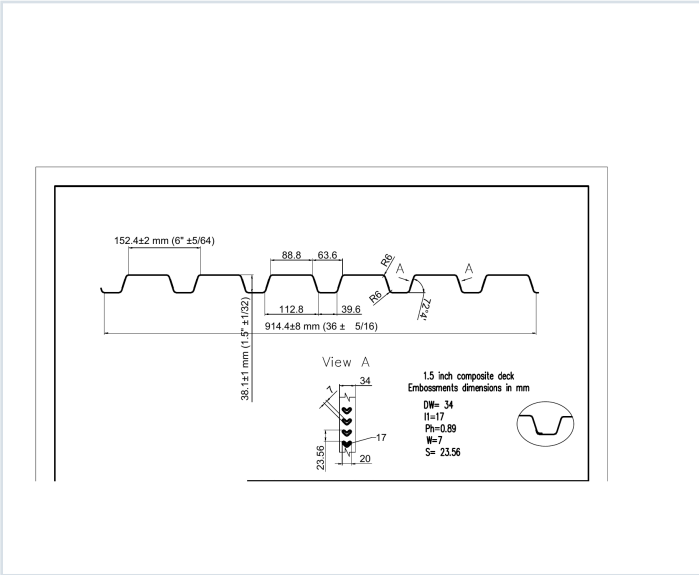
15CD-FY80-18-G40

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 80	18 ga	G40

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.6	60	0.291	0.306	10455	11006	0.252	0.288

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 50, 51, 52, 53
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

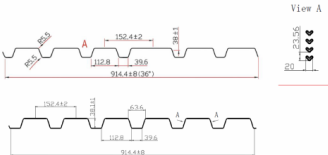
15CD-FY80-18-G40 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 50

GRADE 80 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lbs)	E _x (in ⁴)	S _x (in ³)	S _y (in ³)	S _{xy} (in ⁴)	M _x (in ⁴)	M _y (in ⁴)	I _x (in ⁴)	I _y (in ⁴)
22	0.0395	16	60	0.166	0.175	0.000	60.0	60.0	0.162	0.167
20	0.0368	20	60	0.206	0.215	0.000	7986	7798	0.178	0.209
18	0.0341	24	60	0.246	0.255	0.000	10400	10006	0.192	0.228
16	0.0314	30	60	0.306	0.315	0.000	13440	12806	0.214	0.263

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI S100-2017, AISI S100-2012 and AISI S100-2008.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _n) (kips) One Flange Loading End Bearing	Web Crippling (R _n) (kips) One Flange Loading Interior Bearing
22	2908	985	1056
20	4585	1572	1653
18	6036	2097	2263
16	7463	2677	2880

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI S100-2017, AISI S100-2012 and AISI S100-2008.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	6'-0"	8'-0"	10'-0"	12'-0"	14'-0"	16'-0"	18'-0"	20'-0"	22'-0"	24'-0"	26'-0"	28'-0"	30'-0"
Single	22	197	163	137	117	101	88	77	68	61	55	49			
	20	279	230	194	165	142	124	109	96	86	77	70			
	18	358	297	249	212	183	160	140	124	111	99	90			
Double	22	167	138	116	99	85	74	65	58	52	46	42			
	20	246	203	168	142	122	105	92	81	72	64	58			
	18	323	263	214	183	158	138	121	107	95	84	76			
	16	375	308	252	215	185	161	140	124	110	98	89			
Triple	22	209	175	145	124	107	93	82	73	65	58	52			
	20	296	243	197	165	142	124	109	96	86	77	70			
	18	387	313	255	217	187	163	143	127	113	102	92			
	16	466	388	324	276	238	207	182	161	144	129	116			

Notes:
1. All section properties and ASD flexural strengths are calculated in accordance with AISI S100-2017, AISI S100-2012 and AISI S100-2008.
2. Loads shown in tables are uniformly distributed superimposed loads in psi. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased for moment, clear span dimensions.
3. Working Moment Formula used for beam design: $M = \frac{wL^2}{8}$ for single span, $M = \frac{wL^2}{16}$ for two span, $M = \frac{wL^2}{24}$ for three span or more. M is in ft-kips.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 51

Span	Gage	5'-0"	6'-0"	8'-0"	10'-0"	12'-0"	14'-0"	16'-0"	18'-0"	20'-0"	22'-0"	24'-0"	26'-0"	28'-0"	30'-0"
Single	22	197	163	137	117	101	88	77	68	61	55	49			
	20	279	230	194	165	142	124	109	96	86	77	70			
	18	358	297	249	212	183	160	140	124	111	99	90			
Double	22	167	138	116	99	85	74	65	58	52	46	42			
	20	246	203	168	142	122	105	92	81	72	64	58			
	18	323	263	214	183	158	138	121	107	95	84	76			
	16	375	308	252	215	185	161	140	124	110	98	89			
Triple	22	209	175	145	124	107	93	82	73	65	58	52			
	20	296	243	197	165	142	124	109	96	86	77	70			
	18	387	313	255	217	187	163	143	127	113	102	92			
	16	466	388	324	276	238	207	182	161	144	129	116			

Note:
For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.847.

Construction Span Table – 20 psf Construction Load

Normal Weight Concrete (150 pcf)					Lightweight Concrete (115 pcf)						
Total Slab Depth	Deck Type	Maximum Unstressed Clear Span	1 span	2 span	3 span	Total Slab Depth	Deck Type	Maximum Unstressed Clear Span	1 span	2 span	3 span
16" (1-5/8") 37 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"		400 (1-1/2") 49 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"	
	15x16x20 ga	8' 0"	8' 0"	8' 0"			15x16x20 ga	8' 0"	8' 0"	8' 0"	
	15x16x18 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x18 ga	12' 0"	13' 1"	13' 6"	
	15x16x16 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x16 ga	8' 0"	8' 0"	8' 0"	
	15x16x14 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x14 ga	8' 0"	8' 0"	8' 0"	
400 (1-1/2") 49 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"		450 (1-5/8") 43 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"	
	15x16x20 ga	8' 0"	8' 0"	8' 0"			15x16x20 ga	8' 0"	8' 0"	8' 0"	
	15x16x18 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x18 ga	8' 0"	8' 0"	8' 0"	
	15x16x16 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x16 ga	8' 0"	8' 0"	8' 0"	
	15x16x14 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x14 ga	8' 0"	8' 0"	8' 0"	
450 (1-5/8") 43 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"		500 (1-3/4") 49 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"	
	15x16x20 ga	8' 0"	8' 0"	8' 0"			15x16x20 ga	8' 0"	8' 0"	8' 0"	
	15x16x18 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x18 ga	8' 0"	8' 0"	8' 0"	
	15x16x16 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x16 ga	8' 0"	8' 0"	8' 0"	
	15x16x14 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x14 ga	8' 0"	8' 0"	8' 0"	
500 (1-3/4") 49 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"		550 (1-7/8") 46 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"	
	15x16x20 ga	8' 0"	8' 0"	8' 0"			15x16x20 ga	8' 0"	8' 0"	8' 0"	
	15x16x18 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x18 ga	8' 0"	8' 0"	8' 0"	
	15x16x16 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x16 ga	8' 0"	8' 0"	8' 0"	
	15x16x14 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x14 ga	8' 0"	8' 0"	8' 0"	
550 (1-7/8") 46 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"		600 (1-7/8") 46 PSF	15x16x22 ga	8' 0"	8' 0"	8' 0"	
	15x16x20 ga	8' 0"	8' 0"	8' 0"			15x16x20 ga	8' 0"	8' 0"	8' 0"	
	15x16x18 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x18 ga	8' 0"	8' 0"	8' 0"	
	15x16x16 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x16 ga	8' 0"	8' 0"	8' 0"	
	15x16x14 ga	8' 0"	7' 8"	11' 0"	12' 0"		15x16x14 ga	8' 0"	8' 0"	8' 0"	

Note:
Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY80-18-G40 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 52

22 ga Normalweight Concrete (145 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	37	400	400	375	374	209	232	202
4	37	400	400	400	397	340	293	238
4.5	43	400	400	400	400	400	397	311
5	49	400	400	400	400	400	400	369
5.5	55	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	177	166	138	123	110	98	89	80
4	224	187	175	166	159	125	113	102
4.5	273	241	213	190	170	153	138	125
5	323	289	255	228	202	182	164	148
5.5	375	331	294	262	235	211	191	175
6	400	378	336	300	269	242	218	197

Note
ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

20/18/16 ga Normalweight Concrete (145 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	37	400	400	400	382	327	283	246
4	37	400	400	400	400	400	368	312
4.5	43	400	400	400	400	400	400	381
5	49	400	400	400	400	400	400	400
5.5	55	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	236	181	170	161	136	122	110	100
4	276	242	215	192	172	155	142	127
4.5	333	296	263	236	211	190	172	156
5	390	352	313	280	251	226	205	186
5.5	450	400	364	325	292	263	238	216
6	400	400	400	372	334	301	272	247

Notes
1. Because of the profile of the embossments, there is no gain in strength for the composite deck slab when the deck gets thicker than 20 gage. However, the construction spans do get longer for 18 and 16 gage deck.
2. ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

STEEL CATALOG PAGE 53

22 ga Lightweight Concrete (115 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	400	359	374	260	225	196
4	28	400	400	400	386	329	285	248
4.5	33	400	400	400	400	400	348	304
5	37	400	400	400	400	400	400	362
5.5	42	400	400	400	400	400	400	400
6	46	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	172	152	135	121	108	97	88	80
4	208	183	172	163	158	124	112	102
4.5	267	236	210	188	168	152	137	125
5	318	285	250	224	201	181	164	149
5.5	369	327	291	260	234	211	191	175
6	400	374	333	298	268	242	218	199

Note
ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.

20/18/16 ga Lightweight Concrete (115 pcf, $F_c = 3,000$ psi)

Slab Thickness (inches)	Weight (pcf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	400	400	367	314	272	238
4	28	400	400	400	400	398	346	302
4.5	33	400	400	400	400	400	400	370
5	37	400	400	400	400	400	400	400
5.5	42	400	400	400	400	400	400	400
6	46	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	279	181	161	147	132	120	108	99
4	306	235	209	187	169	152	136	125
4.5	325	288	257	230	207	187	169	154
5	368	344	306	275	247	223	202	184
5.5	400	400	357	320	288	260	236	215
6	400	400	400	366	330	298	271	246

Notes
1. Because of the profile of the embossments, there is no gain in strength for the composite deck slab when the deck gets thicker than 20 gage. However, the construction spans do get longer for 18 and 16 gage deck.
2. ANSI/AISC C-307 permits the use of Grade 80 steel for composite deck, but it limits the yield strength for determining composite deck slab strength to 50 ksi. Therefore for Grade 80 steel, 50 ksi values are used.